GPU Task-Graph Scheduling Evaluation

Measured results for six KESCO-derived applications

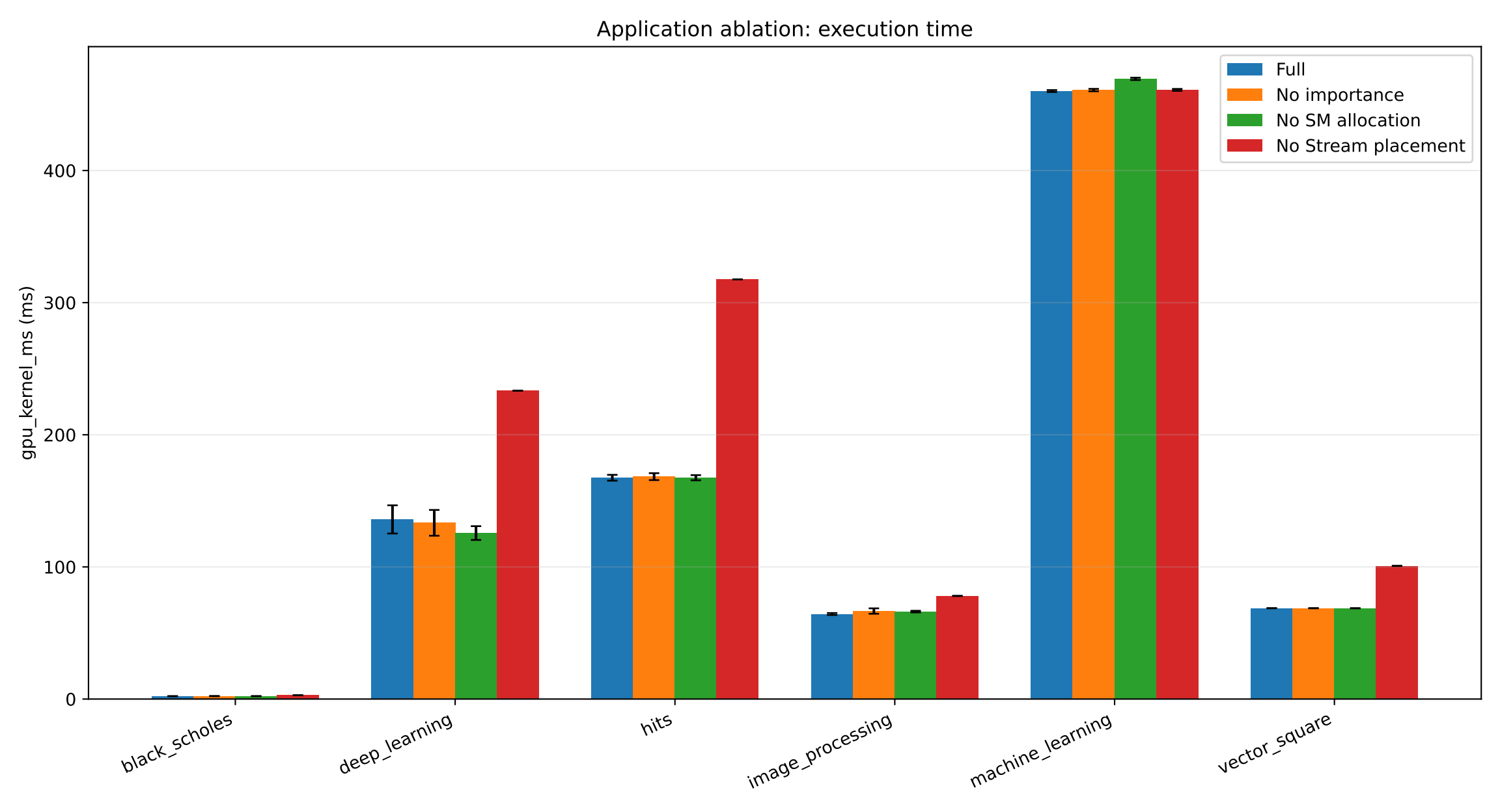
Experimental design
• 6 applications, 4 staged ablation variants
• corrected ablation measurements; warmup policy is recorded in the CSV
• mean and 95% confidence interval reported
• Fixed three-Stream comparison; numerical validation passed for all 24 conditions

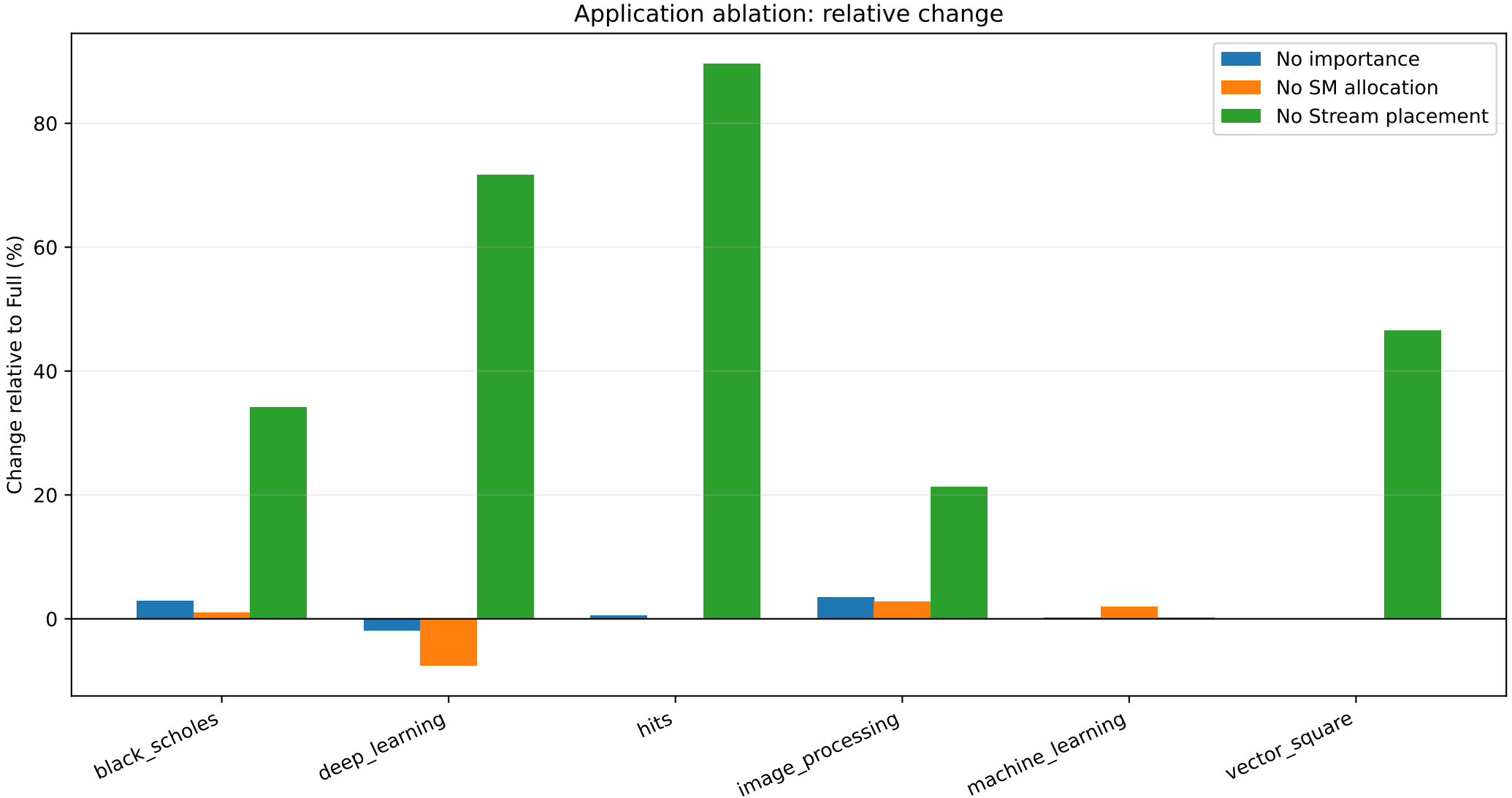
Variants
Full: importance-aware Stream placement and SM allocation
No importance: same pipeline with bottom-level priority disabled
No SM allocation: same Stream count with equal SM allocation and no Green Context
No Stream placement: one execution Stream

Interpretation
Stream placement is the most consistently influential component. Removing it increases execution time substantially for deep_learning, hits, and vector_square. The effect of importance and SM allocation is workload-dependent.

Timing note
gpu_kernel_ms is measured using CUDA Events from the first task start to the last task completion on the GPU.

Source: results/application_ablation_corrected/


Application ablation: execution time
gpu_kernel_ms (ms)
0
100
200
300
400
Full
No importance
No SM allocation
No Stream placement
black_scholes
deep_learning
hits
image_processing
machine_learning
vector_square


Application ablation: relative change
No importance
No SM allocation
No Stream placement
Change relative to Full (%)
80
60
40
20
0
black_scholes
deep_learning
hits
image_processing
machine_learning
vector_square